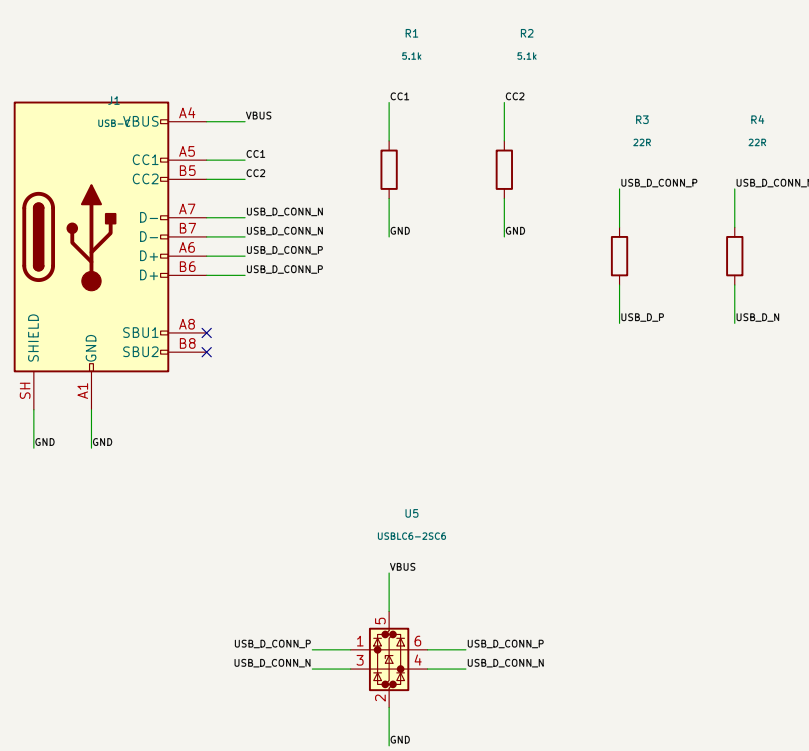
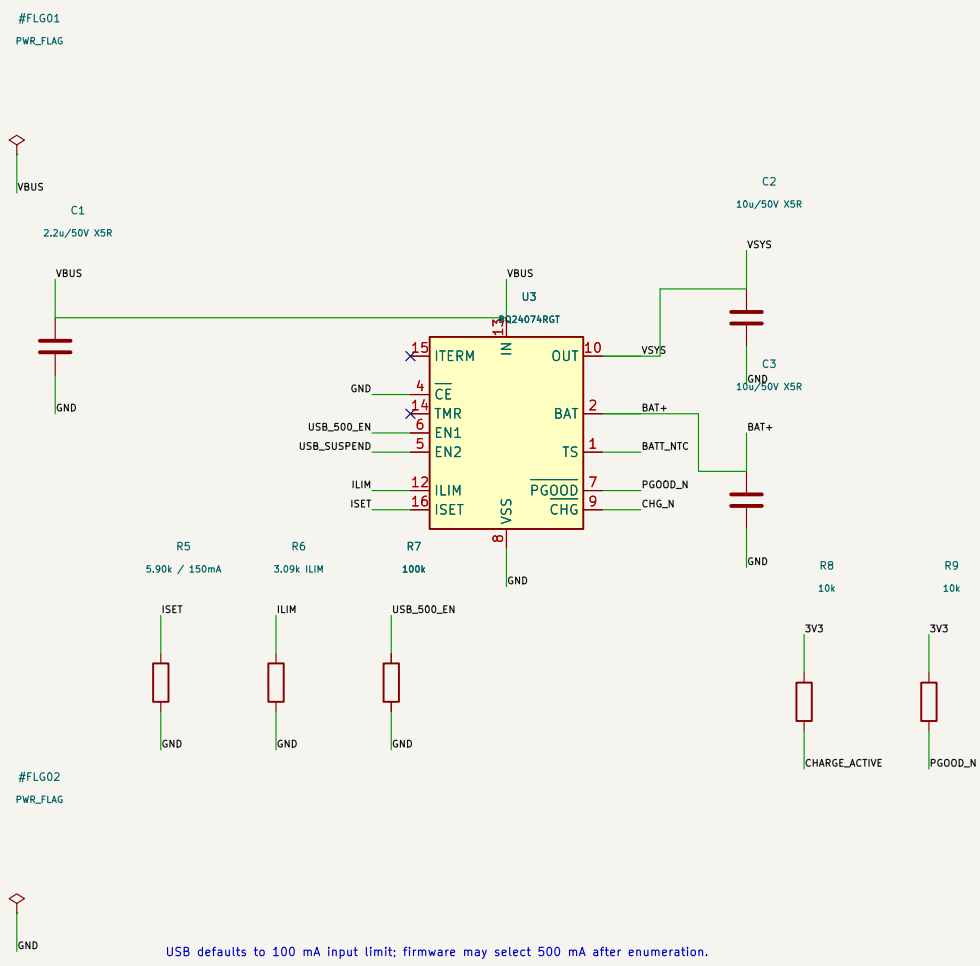


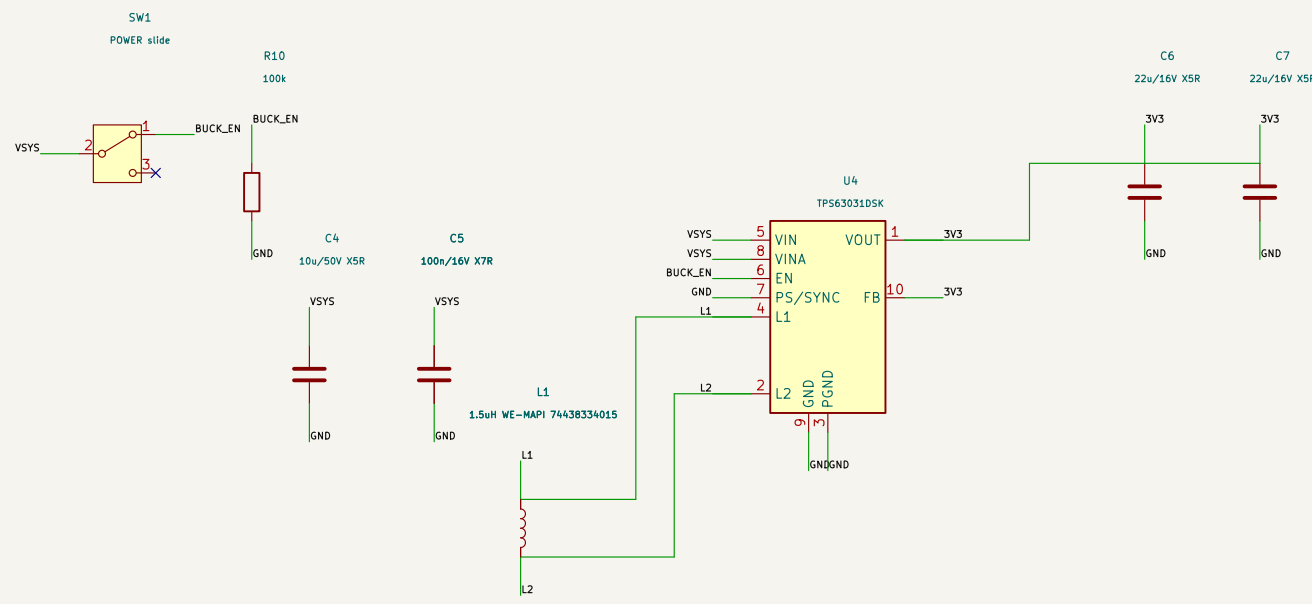
USB-C / 5 V INPUT / CC / DATA



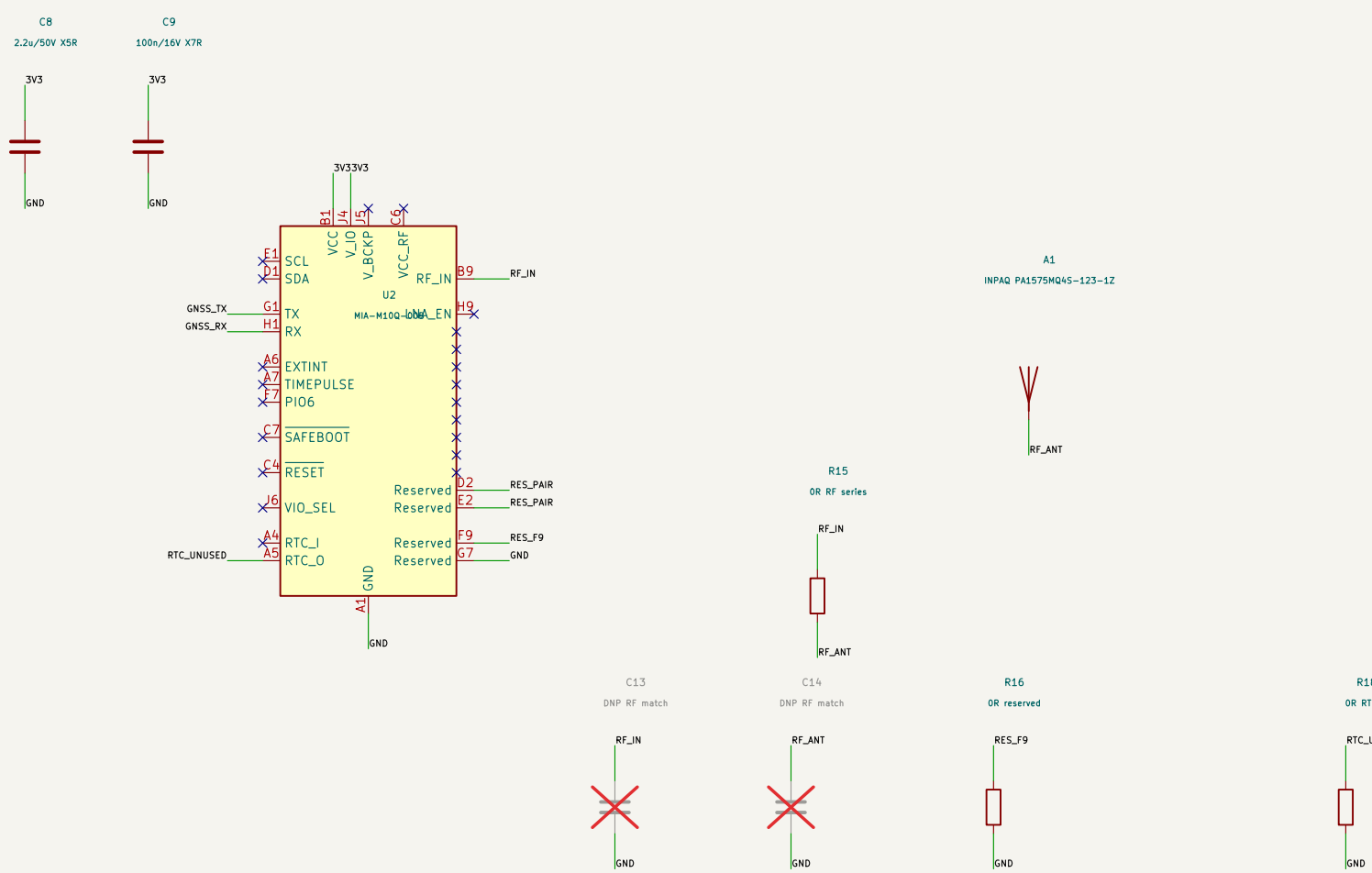
1S LI-ION CHARGE + POWER PATH



3.3 V BUCK-BOOST



MIA GNSS + DIRECT CERAMIC PATCH / RF MATCH

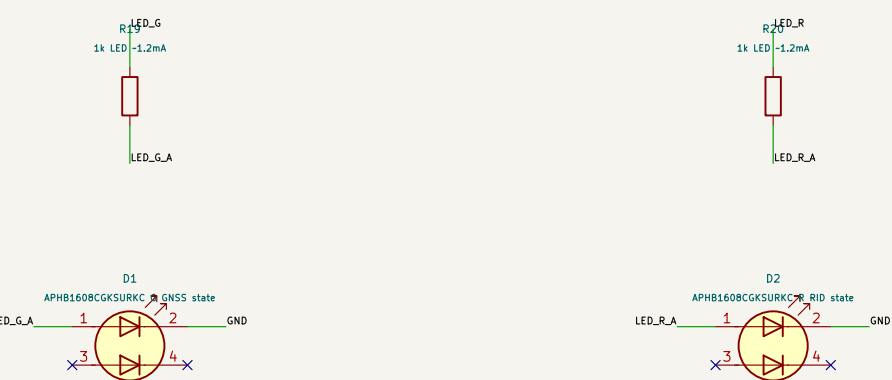


1S 1S isolated path + NTC



Protected 1S pack with 10k NTC required: charge ~150 mA is provisional.

STATUS INDICATORS / ISOLATED CHARGE SENSE



G: GP06, blink acquiring / steady valid GNSS. R: GP07, actual successful R/D transmit state.

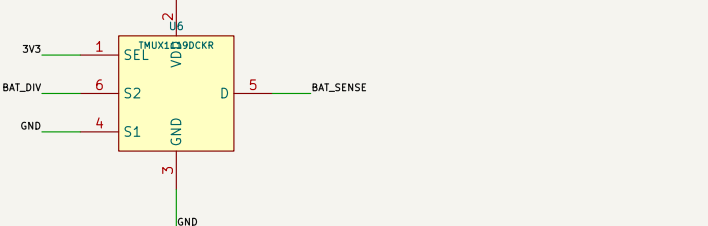
S green: 3V3 system rail enabled (not firmware-health proof). S red: charger CHG_N low, independent of MCU power. Charging can show amber while running.

Q1 prevents the VBUS-powered CHG mode from driving an unpowered ESP input. Charging = (PGOOD_N low) AND (CHARGE_ACTIVE high); ignore CHARGE_ACTIVE without valid USB input.

G/R indicators require future firmware. An illuminated R is not proof of compliant RF/R/D transmission.

POWER-OFF ADC ISOLATION / USB POLICY / RECOVERY STRAP

Native CAD checks passed; battery, sourcing, assembly and physical qualification pending.



VB6 SEL=3V3, ON -> ADC grounded; on -> BAT_DIV, C16 stays on divider side. Verify rail-collapse sequencing.

USB EN2/EN1: 00=100mA, 01=500mA, 11=suspend. Enter EN1 first; resume EN2 first. Avoid 10 ILIM state.

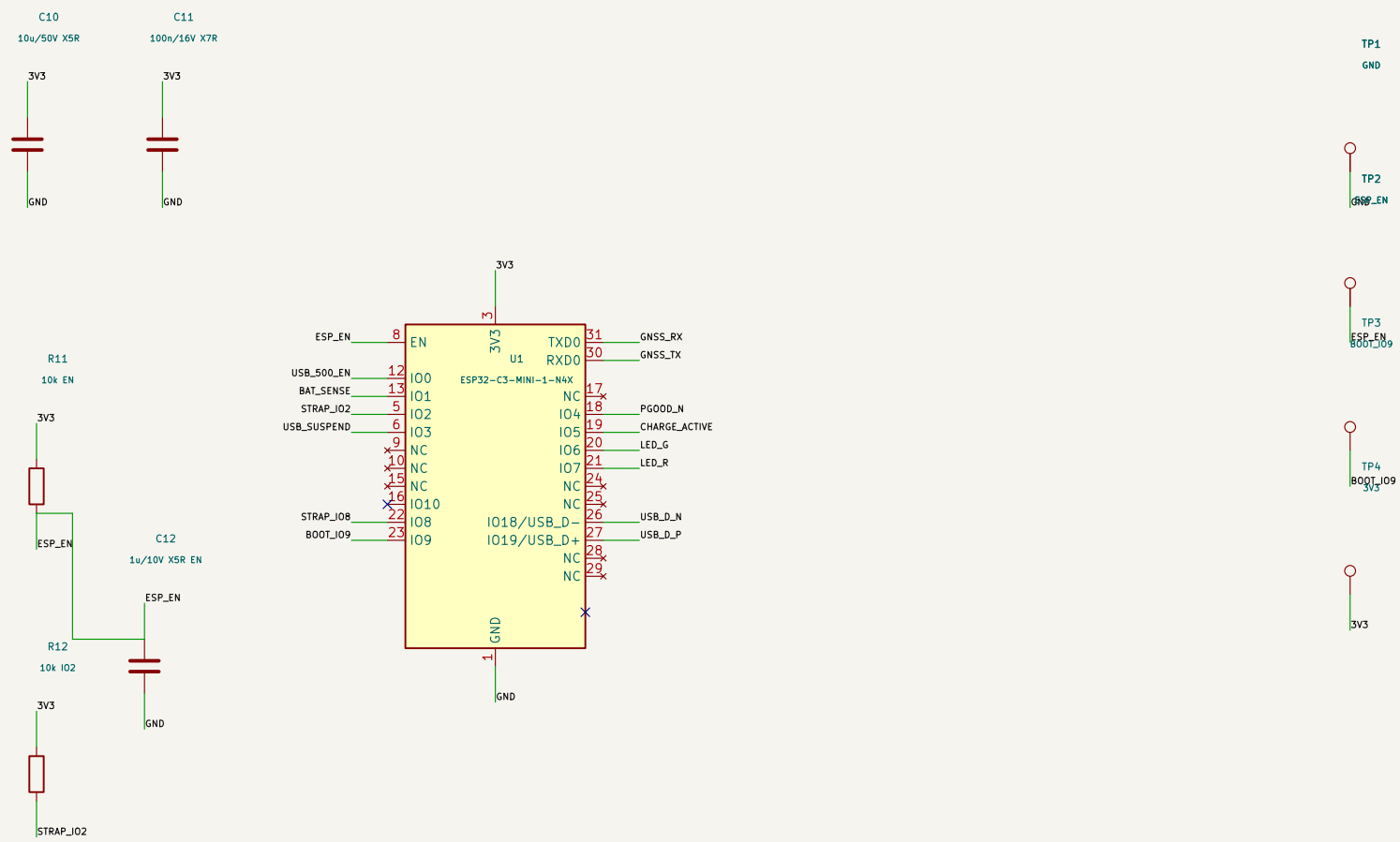
Default 100mA. S0F/connected is NOT a configured 500mA grant. Validate host grant before selecting USB500.

MCU-off charging cannot detect host suspend. USB power policy and prototype measurements remain release gates.

GP08 pull-up supports manual download with GP08/BOOT low. No status LED on strapping pins.

Charger EN2 GP03. S-green shows enabled system rail; G/R use GP06/GP07 firmware state.

ESP32-C3 BLE/WI-FI + PROGRAMMING



ESP antenna is inside overall outline with board-corner recess and all-layer keepout.

CAD CHECKED: BATTERY / RF / SOURCING GATES - NO FABRICATION RELEASE